

Optimization of a Standalone VR Experience through User Perception: A Journey to the Hidden Spaces of the Burgos Cathedral

David Checa[†]
Universidad de Burgos

Mario Alaguero*
Universidad de Burgos

ABSTRACT

This project offers a groundbreaking virtual reality (VR) experience that allows visitors to explore inaccessible areas of the Burgos Cathedral. Employing photogrammetry techniques, these spaces have been digitally recreated in high detail. The application, optimized for standalone VR devices, has undergone meticulous optimization. This was achieved by analyzing visitor attention patterns, enhancing resolution in frequently observed areas to ensure a rich and seamless experience. This initiative not only significantly improves the visitor experience but also serves as a model for the dissemination of cultural heritage through VR.

Keywords: Virtual Reality, Photogrammetry, Game for Learning, Storytelling.

Index Terms: I.3.7 Three-Dimensional Graphics and Realism.

1 INTRODUCTION

The Burgos Cathedral, a remarkable example of Gothic architecture, stands as a testament to medieval artistry and craftsmanship. Recognized by UNESCO for its historical and cultural significance, the cathedral houses several sections hidden from public view, rich in untold stories and architectural wonders [1]. These parts are hidden due to their inaccessibility, which makes them impossible to see by conventional means. This project attempts to overcome these physical barriers through virtual reality (VR), providing a realistic exploration of the cathedral's secret and inaccessible spaces, thus seamlessly connecting historical heritage with contemporary technology. The combination of virtual reality (VR) and photogrammetry has become increasingly popular for visualizing and preserving cultural heritage objects [2]. Photogrammetry offers a reliable method for accurate image-based modeling of real-world objects, while VR applications not only provide visualization but also an immersive and interactive experience. This synergy is particularly effective in cultural heritage applications, where it's crucial to represent the objects or environments with high fidelity [3]. The process involves the integration of high-quality photogrammetric 3D models with virtual reality technologies, along with advanced visualization engines to create nearly real-world experiences [4]. This project is driven by two core objectives: firstly, to democratize access to the cathedral's otherwise inaccessible areas, thereby enriching the narrative of Burgos Cathedral for visitors worldwide. Secondly, it aims to pioneer the use of innovative autonomous VR technology combined with photogrammetry in

the field of cultural heritage. In doing so, the project not only offers an enhanced visitor experience, but also aims to establish new solutions to optimize these experiences and transform the way we interact with and understand historic sites.

2 METHODOLOGY

The methodology of the project involved a comprehensive approach that started with visiting and selecting the most remarkable and publicly inaccessible areas within the cathedral. This selection process focused on identifying spaces that best represent the marvel of the cathedral to the public, utilizing the potential of virtual reality to create an immersive and meaningful experience of presence. Once the specific spaces were carefully selected, the next phase involved capturing aerial imagery using drone technology to accurately gather detailed architectural features of the cathedral. The next step was the transformation of this imagery into detailed 3D models through photogrammetry, ensuring high fidelity to the actual structure. These models were then optimized for VR through retopology, balancing detail with performance. Subsequently, the models were integrated into a game engine, and further refined based on user gaze data, adjusting texture resolutions in areas of high interest while reducing them in less observed regions, to manage application performance while maintaining visual quality.

3 PHOTOGAMMETRY

For photogrammetry, the project utilized a dual approach: aerial images taken by drones and ground-level photographs captured from the cathedral's roofs and higher sections. This combination ensured comprehensive coverage of the structure. The drone images provided a broad view, while the ground-level photos offered detailed perspectives of intricate architectural elements. This dual-source approach was crucial for capturing the full architectural complexity of the Burgos Cathedral, ensuring the VR experience was as authentic and detailed as possible. These images were then processed in photogrammetry software Reality capture, creating a highly accurate and detailed 3D model as can be seen in figure 1.

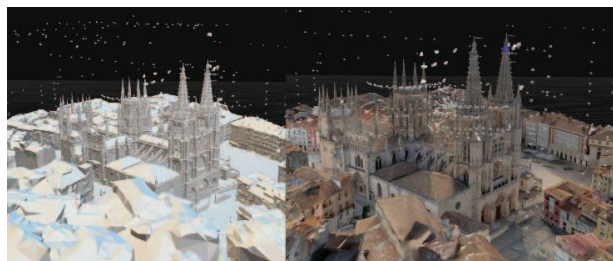


Figure 1: Highly accurate and detailed photogrammetry 3D model.

This highpoly model was then strategically segmented into various parts based on their visual significance within the scene.

[†]e-mail: dchecha@ubu.es

* e-mail: malaguero@ubu.es

Each segment received a different polygon count, with more intricate areas having higher polygon densities. This approach allowed for detailed rendering of key features while managing overall resource demands. The total polygon count for each entire scene was carefully kept under one million polygons.

4 OPTIMIZATION

In the optimization process for the photogrammetric model of the Burgos Cathedral for Meta Quest 2, several key steps were undertaken. Initially, the 3D models underwent polygon reduction, focusing on retaining intricate details in areas of interest while simplifying other sections. Texture optimization was crucial, involving the compression and resizing of textures to balance visual quality with file size. This was paired with meticulous adjustments to the mesh, smoothing out imperfections and enhancing its structure. Additionally, any defects in the textures were rectified using Substance Painter. Once optimized, the models were integrated into Unreal Engine. Level of Detail (LOD) models were created, allowing the system to display simpler models from a distance and more detailed ones up close. To enhance performance, the number of materials and shaders was minimized, using more efficient, less resource-intensive options. Occlusion culling was employed to prevent rendering of objects not in the player's view, conserving processing resources. Also, asset streaming was used to dynamically load parts of the scene. All elements used unlit textures to save on lighting calculation costs. Profiling tools were employed throughout to analyze performance and pinpoint any bottlenecks in the VR experience. The entire optimization process was iterative, with regular testing on the Meta Quest 2 to identify and address performance issues.

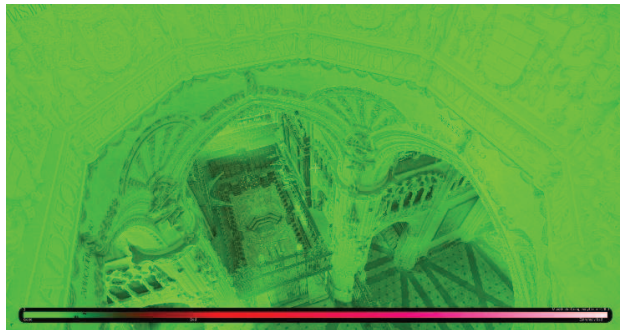


Figure 2: Shader Complexity Analysis in a Given Scene.

User testing then commenced, focusing on what users paid attention to within the VR environment. The user testing for the VR experience was conducted using Oculus Quest Pro glasses, which feature eye-tracking technology. This allowed for precise data collection on where users were looking during the VR sessions. Our custom-built framework [5] was instrumental in this process, as it is capable of recording these eye-tracking data. The collected data was then used to generate heatmaps, visually representing user focus areas within the virtual environment. These heatmaps played a crucial role in the iterative refinement process, informing decisions on where to allocate higher polygon counts and texture resolutions in the VR models. This additional iteration was pivotal in fine-tuning the VR experience to ensure optimal user engagement and visual fidelity.

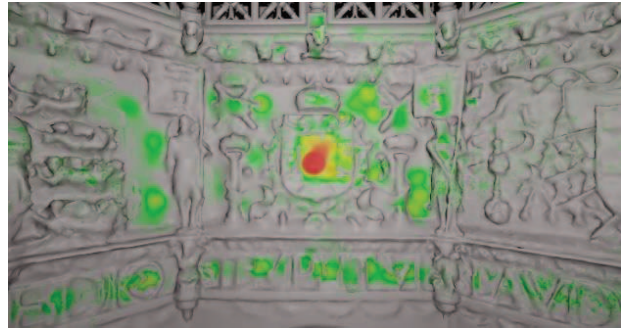


Figure 3: Heatmap Example from a User Session.

5 DISCUSSION

This project demonstrates a practical application of virtual reality in cultural heritage. Its integration in the Burgos Cathedral, which receives approximately 40,000 visitors monthly, indicates a significant potential for VR in cultural sites. A setup has been installed in one of the cathedral's chapels, equipped with ten VR headsets. Every quarter of an hour, groups of ten visitors are given the opportunity to experience the VR tour. The headsets continue to collect data on where visitors focus their attention. This ongoing data collection is planned to be used for iterative improvements of the VR experience. Future enhancements may include personalized content delivery, where the VR tour adapts in real-time to focus on areas of highest interest to individual users. Additionally, this data will aid in better understanding visitor engagement patterns, which can inform future expansions of the VR experience to other parts of the cathedral or similar historical sites. This approach not only enriches visitor experience but also provides valuable insights for continuous refinement of the VR application in cultural heritage settings.

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